

FIFTH SEMESTER DIPLOMA EXAMINATION IN CIVIL ENGINEERING

QUANTITY SURVEYING II & VALUATION**MODEL QUESTION PAPER**

[Time: 3 hours]

(Maximum marks: 100)

PART—A

(Maximum marks: 10)

Marks

I. Answer the following questions in one or two sentences.

- a) How the length of the common rafter shall be calculated in a tiled roof having $1/3$ span.
- b) The estimate of a multistoried building is to prepared floor wise. Why?
- c) A slab culvert has a clear span of 2m with bearing of slab 50 cm on either side. What shall be the length of hand rails on each side.
- d) Write down the standard units
1) R C C for slab 2) Reinforcement for R C C
- e) Define sinking fund

(5 x 2=10)

PART—B

(Maximum marks: 30)

(Answer any five questions)

II a) Work out the quantities for the following items for the septic tank in fig (I)

- 1) Earth work excavation. 2) P C C 1:4:8 for foundation.
- b) Work out the quantity of brick masonry in C M 1:6 for the septic tank in fig (I)
- c) For the slab culvert given in fig (II), work out the quantity of earth work excavation for abutment and wing wall
- d) Work out the quantity P C C 1:4:8 for foundation of abutment and wing wall for the slab culvert in fig (II).
- e) Write down the detailed specification for brick masonry in C M 1:6 for super structure
- f) What are the various methods of calculating depreciation of a property. Briefly explain any one method.

g) What is bar bending schedule? Draw typical bar bending

(5 x 6=30)

PART—C

(Maximum marks: 60)

(Answer one full question from each module)

MODULE—I

- III. a) The attached fig (III) shows the details of building, work out the quantities of earth work excavation for foundation 8
- b) Work out the quantity of random rubble masonry for foundation and basement 7

OR

- IV. For the building given in fig (III), work out the quantity of brick masonry in C M 1:6 for superstructure. 15

MODULE—II

- V. a) Work out the quantity of random rubble masonry in C M 1:6 for the slab culvert shown in fig (II) 8
- b) Work out the quantity of R C C 1:1½:3 for the deck slab of culvert shown in fig (II) 7

OR

- VI. a) Work out the quantity of a R.C.C 1:2:4 for cantilever retaining wall of 20 m length as shown in fig: IV 8
- b) Work out the quantity of Earth work excavation in ordinary soil for foundation of fig: IV 7

MODULE--III

- VII. Prepare the bar bending schedule for the cantilever retaining wall shown in fig IV 15

OR

- VIII a) Write down the detailed specification for earth work excavation in ordinary soil for foundation. 8

- b) Write down the detailed specification for R.C.C 1:2;4 using 20 mm nominal size aggregate for slab

7

MODULE—IV

- IX. A building is constructed at a cost of Rs.15,00,000/- on a land purchased at Rs.5,00,000/-
The owner of property expect a return of 9% on cost of construction and 10% on cost
Of land .The building is estimated to a life of 60 years at the end of which it requires
Rs. 90,00,000/- for the construction of new building in its place.
Determine the standard rent if the property given
- 1.Rate of interest for sinking fund 8%
 2. Annual repairs @2% of the cost of construction
 3. all other outgoings 28% of the net income of the property
 - 4.Scrap value at the end of the usefull life of the building as 10%
- OR
- X .a) List the factors affecting the value of the property
- b) Workout the value of years purchase for an old building for its future life is 17 years and
the rate of interest is 7.5% on capital and 4%for sinking fund

15

8

7

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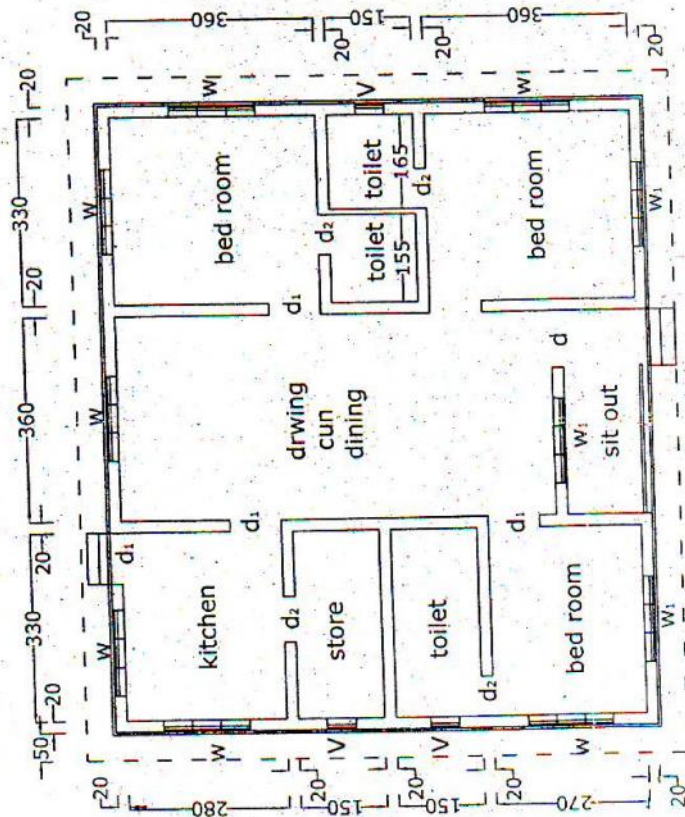
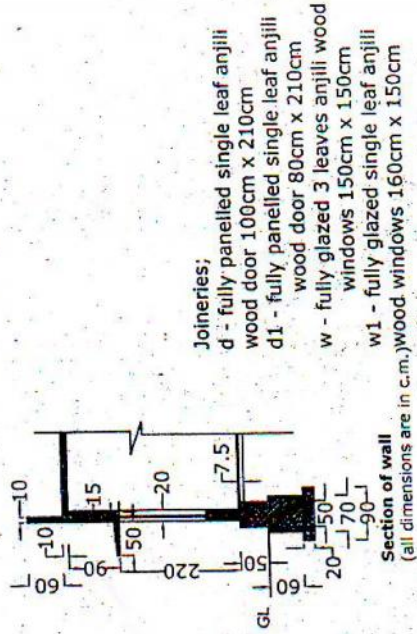


Fig. III Plan of building
(all dimensions are in cm)



Joineries;
d - fully panelled single leaf anjili
wood door 100cm x 210cm
d1 - fully panelled single leaf anjili
wood door 80cm x 210cm
w - fully glazed 3 leaves anjili wood
windows 150cm x 150cm
w1 - fully glazed single leaf anjili
wood windows 160cm x 150cm

Section of wall
(all dimensions are in cm)

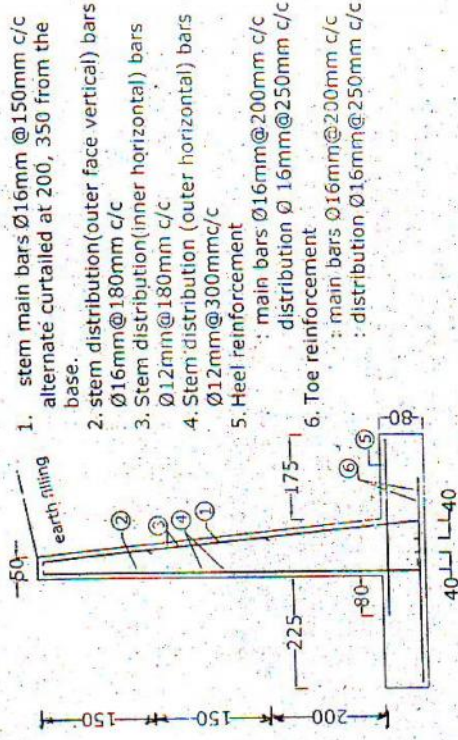


Fig. IV Section of a RCC retaining wall
(all dimensions are in cm)

1. stem main bars $\phi 16\text{mm}$ @ 150mm c/c alternate curtailed at 200, 350 from the base.
2. stem distribution (outer face vertical) bars $\phi 16\text{mm}$ @ 180mm c/c
3. Stem distribution (inner horizontal) bars $\phi 12\text{mm}$ @ 180mm c/c
4. Stem distribution (outer horizontal) bars $\phi 12\text{mm}$ @ 300mm c/c
5. Heel reinforcement
: main bars $\phi 16\text{mm}$ @ 200mm c/c
distribution $\phi 16\text{mm}$ @ 250mm c/c
6. Toe reinforcement
: main bars $\phi 16\text{mm}$ @ 200mm c/c
distribution $\phi 16\text{mm}$ @ 250mm c/c